

Vision-Based Autonomous Navigation for Unmanned Ground Vehicle

Detailed Technical Workflow — Software Solution

Problem Statement ID	26126
Title	Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment
Organization	Bharat Electronics Limited
Category	Software
Theme	Smart Automation

1. Background & Problem Understanding

Outdoor Unmanned Ground Vehicles (UGVs) operate in environments where terrain is unpredictable, lighting conditions change constantly, and GPS signals are often unreliable or unavailable. To achieve genuine autonomy in applications such as search-and-rescue, agriculture, and delivery, a UGV must rely on onboard computer vision as its primary means of perceiving and understanding its surroundings.

The objective of this problem statement is to design an autonomous navigation system for a UGV operating in a GPS-denied outdoor environment, using camera feeds as the primary sensor. The solution must address three coupled challenges:

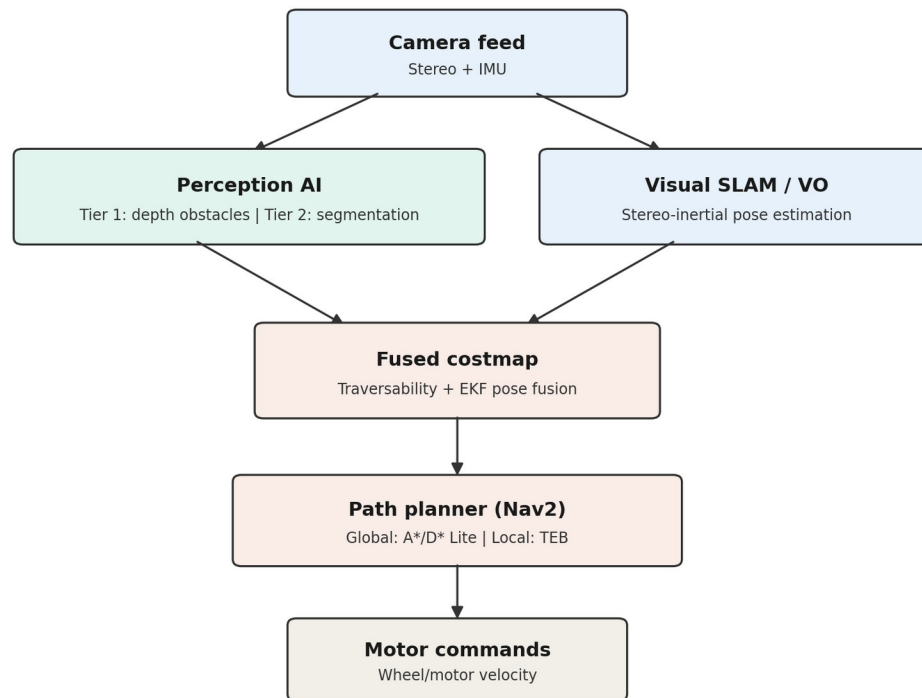
- Path Detection — real-time identification of safe, traversable paths versus hazards such as rocks, ditches, and trees.
- Visual Localization — estimating the UGV's position and orientation without GPS, using visual data alone.
- Collision Avoidance — dynamically routing the vehicle around sudden obstacles while continuing toward its destination.

Success is defined as collision-free navigation from a designated Point A to Point B, across varied outdoor scenarios, using a single integrated software module.

2. Proposed Solution — Overview

The proposed system is a closed-loop pipeline in which a stereo depth camera with an integrated IMU feeds two parallel processing branches — a perception branch that builds a traversability map, and a visual SLAM/odometry branch that estimates the vehicle's pose. Both outputs are fused into a single costmap that drives a standard path-planning stack, which in turn issues motor commands. This mirrors, and is built on top of, the ROS2 Nav2 navigation framework, adapted to consume vision-based inputs in place of lidar.

3. System Architecture



4. Perception Module — Path & Obstacle Detection

4.1 Tier 1 — Depth-based obstacle detection

- Uses the stereo depth map directly; no machine learning required.
- Any surface point above a configurable ground-height threshold, within the vehicle's path, is flagged as an obstacle.
- Runs at full frame rate on modest compute, and acts as a guaranteed safety-net layer even if the learned model underperforms on unfamiliar terrain.

4.2 Tier 2 — Learned terrain segmentation

- A lightweight semantic segmentation model (Fast-SCNN or BiSeNetV2) classifies terrain into safe versus hazardous categories (grass, packed dirt, mud, ditch, dense brush).
- Trained on the RUGD or RELIS-3D off-road datasets, fine-tuned on site-specific footage where available.
- Exported to ONNX/TensorRT for real-time inference.
- Distinguishes cases depth alone cannot — e.g. a flat mud patch and flat solid ground sit at the same height but are not equally traversable.

Both tiers write into a single shared traversability cost layer, with Tier 1 acting as a hard safety floor and Tier 2 adding semantic nuance on top.

5. Visual Localization Module

- Stereo-inertial ORB-SLAM3 or VINS-Fusion performs pose estimation, using the stereo baseline for metric scale and the IMU to maintain tracking through brief visual dropouts.
- An Extended Kalman Filter (via the robot_localization package) fuses the SLAM pose estimate with wheel odometry, so no single sensor's failure mode brings down localization entirely.
- Loop closure is enabled to correct accumulated drift when the vehicle revisits previously seen terrain.

Pure monocular SLAM is deliberately avoided: without a stereo baseline or IMU, monocular visual odometry has no absolute scale reference and is prone to compounding scale drift over long outdoor runs — a risk this design removes structurally rather than compensating for later.

6. Path Planning Module

- Built on the ROS2 Nav2 stack, with the lidar-based costmap input replaced by the fused perception output described above.
- An inflation layer is applied around every obstacle cell, giving the vehicle an automatic clearance margin.
- Global planning uses A* or D* Lite over the traversability grid to determine the overall route to the destination.
- Local planning uses the Timed Elastic Band (TEB) planner, chosen because it accounts for the vehicle's real turning kinematics and reacts to obstacles that appear suddenly during motion.

7. Software-Only Build Plan (Simulation First)

As hardware is not yet involved, the entire pipeline is first developed and validated in simulation, so that every component can be integration-tested before any physical build begins.

Stage	Focus
Environment setup	Gazebo world (off-road terrain) with a UGV model carrying a simulated stereo camera and IMU plugin.
First closed loop	Nav2 driven purely by Tier 1 (depth-only) costmap — establishes a working Point A to B run with no ML dependency.
Localization integration	Stereo-inertial SLAM/VIO added, fused with simulated wheel odometry via EKF, replacing any placeholder localization.
Perception Tier 2	Segmentation model trained and integrated as a second costmap layer, merged with Tier 1.
Stress testing	Harder simulated scenarios — slopes, denser obstacles, sudden obstacle spawns — used to tune the local planner and inflation radius.

All model training and simulation runs on available development hardware (GPU-equipped laptop); the resulting nodes and pipeline are hardware-agnostic and designed to port to embedded compute for physical deployment at a later stage without architectural changes.

8. Key Differentiators

- Two-tier perception rather than a single model handling everything — a geometry-based safety floor (Tier 1) ensures the system fails safely even if the learned model (Tier 2) misclassifies unfamiliar terrain.
- Stereo-inertial localization chosen deliberately over monocular SLAM, removing the scale-drift problem at the architecture level instead of patching it after the fact.
- Reuse of the mature Nav2 planning stack rather than building a custom planner from scratch, inheriting years of tested robustness.
- Simulation-first validation, catching integration issues before any hardware risk is introduced.
- An explicit degradation plan — a depth-only navigation mode remains a complete, demonstrable fallback if the learned perception layer is not fully mature by evaluation time.
- Sensor fusion (SLAM + wheel odometry + IMU via EKF) treated as a core design choice rather than an afterthought, so no single sensor's momentary failure cascades into total localization loss.

9. Expected Outcome

A functional software module demonstrating collision-free navigation from a designated Point A to Point B in simulated outdoor scenarios, consisting of an integrated perception AI, a visual SLAM/odometry pipeline, and a path planner — directly satisfying the success criteria defined in the problem statement, with a clear, low-risk path to physical hardware deployment thereafter.